React – Fundamentals

Session – 13

1. Run npm run build in your React project folder and verify that a build folder is created with optimized production files.

Ans.

Steps:

- **Open Terminal in your React project.**
- **Run:**

```
npm install
npm run build
```

Output:

```
Creating an optimized production build...

Compiled successfully.

File sizes after gzip:
...
```

2. Sign up for a free Netlify account, connect your GitHub account, and deploy a simple React app (not the one built in class) from a new GitHub repository using Netlify's dashboard.

Hint: Make sure your repository has the build folder generated by npm run build before deploying.

Ans.

Step 1: Create a React App:

```
npm create vite@latest my-music-player
```

Initialize Git.

```
git init

git add .
git commit -m "Initial commit"
```

Create a new GitHub repository and push it.

```
git remote add origin https://github.com/yourusername/my-music-player.git

git push -u origin main
```

3. Update the homepage field in your package.json to a custom URL (e.g., https://yourname-musicplayer.netlify.app), rebuild your React app using npm run build, and redeploy to Netlify to verify that routes work correctly.

Hint: This helps avoid routing issues in SPAs when deployed.

Ans.

package.json:

```
{
  "homepage": "https://yourname-musicplayer.netlify.app"
}
```

Rebuild

```
npm run build
```

Push changes.

```
git add .
git commit -m "Updated homepage"
git push
```

4. Add a new feature to your deployed React app, such as a dark mode toggle or a new About page, then redeploy the updated build to Netlify and confirm the changes are live.

Ans.

```
import { useState } from "react";

function App() {
  const [dark, setDark] = useState(false);

  return (
    <div
      style={{
        background: dark ? "#222" : "#fff",
        color: dark ? "#fff" : "#000",
        minHeight: "100vh",
        padding: "20px",
      }}
```

```
    >
      <button onClick={() => setDark(!dark)}>
        Toggle Dark Mode
      </button>

      <h1>Music Player</h1>
    </div>
  );
}

export default App;
```

Commit and push.

```
git add .
git commit -m "Added dark mode"
git push
```

5. After deploying your React app to Netlify, share the live URL with a friend and ask them to open it on their phone. Collect their feedback on loading speed and appearance, and write two changes you would make to improve user experience.

Ans.

feedback:

- **Loading Speed: Fast, page opened within 2–3 seconds.**
- **Appearance: Clean layout, but buttons could be larger on mobile.**

Two improvements:

- **Increase button size and spacing for easier touch interaction.**
- **Optimize images and enable lazy loading to reduce page load time.**